Grasping via Online Adaptation

Advanced Deep Learning for Robotics [CIT433027]

Hannes Ill
Matr. No. 03786888
hannes.ill@tum.de

Lucas Aschenbach
Matr. No. 03710376
lucas.aschenbach@tum.de

***Abstract*—We study test-time ("online") optimization of grasp configurations guided by a learned grasp quality (GQ) network. From a dataset of 15,547 objects with 480 grasp-configuration/score pairs per object (7.5M grasps total), we train a compact 3D convolutional encoder over signed distance fields (SDFs) and fuse the latent with a 19-D grasp parameterization to predict ϵ-quality scores. Naïve gradient ascent on grasp parameters led to adversarial, invalid grasps. We regularize optimization by (i) a learned normalizing flow prior over the 19-D grasp space and (ii) a differentiable kinematics loss that attracts fingertips to the object surface while penalizing collisions via the SDF. The resulting online optimizer produces valid, contact-rich grasps with substantially fewer penetrations. Our final predictor attains an MSE of 6.08 (vs. 13.44 initially) despite the discontinuity of ϵ-quality, and yields an approximate grasp ordering useful for optimization. We report ablations for grasp quality predictor architectures and discuss remaining failure modes related to non-watertight meshes and sparse collision points.**

I. Introduction

Robust robotic grasping often couples object understanding with search over high-dimensional hand configurations. While offline datasets can teach a notion of grasp quality, practical deployment benefits from *online adaptation*: refining grasp parameters on a specific, newly observed object at test time. We propose a simple pipeline: encode the object as a low-resolution SDF, predict grasp quality for a 19-D hand configuration, and optimize the grasp via gradient ascent through the network. To prevent the optimizer from leaving the manifold of plausible grasps and to encourage physical realism, we introduce a learned grasp prior and a differentiable kinematics loss derived from the SDF.

Contributions: (1) A compact 3D-CNN grasp quality network that generalizes across 15k meshes. (2) A test-time optimization objective combining predicted quality, a normalizing-flow grasp prior, and an SDF-based kinematics term.

II. Related Work

a) Learning-based grasp scoring.: Large synthetic datasets with analytic labels enable grasp-quality predictors that support fast planning, e.g., Dex-Net 2.0 for parallel-jaw grippers [1]. For dexterous hands, several works *optimize* grasp parameters by differentiating through a learned objective: Van der Merwe *et al.* pair a success model with SDF-based collision handling [2], while Liu *et al.* backpropagate through forward kinematics and collision terms for high-DoF grippers [3].

b) Two-stage pipelines and ε-quality.: Closest to our setup, Winkelbauer *et al.* use a two-stage architecture that proposes palm poses and regresses finger configurations along with an ε-quality score from volumetric object encodings [4]. This score describes the smallest amount of a disturbing force in any direction that would break the grasp. We also rely on ε-style supervision, but perform *online* refinement via gradient ascent through a learned quality surrogate.

c) Implicit geometry for contact.: Continuous SDF representations provide gradients useful for contact promotion and collision penalties; DeepSDF is a prominent example [5]. In our case, fixed SDF volumes (from meshes) supply efficient distance queries and normals that regularize optimization.

d) Positioning.: Our approach combines (i) a volumetric 3D-CNN grasp-quality model, (ii) gradient-based refinement as in differentiable planners [2], [3], and (iii) SDF-based geometry, complementing two-stage predictors grounded in ε-quality [4].

III. Dataset and Preprocessing

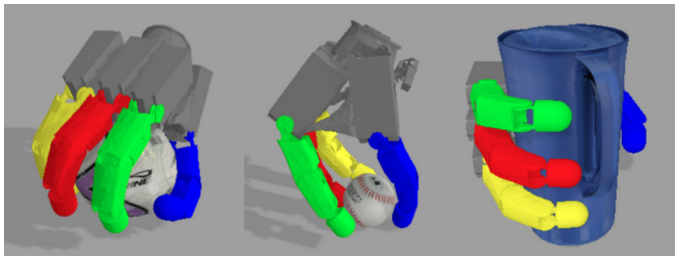


Fig. 1. The best grasps for three scenes from the dataset.

We use a dataset of 15,547 objects, each associated with 480 pairs $(\mathbf{g}, q)$ where $\mathbf{g} \in \mathbb{R}^{19}$ is a grasp configuration and q is the ϵ-quality score. The robot hand has 18 DoF, 6 for the hand-pose and 3 for each of the four fingers. The grasp is represented as a 19-dimensional vector as the hand rotation is expressed in terms of quaternions. Fig. 1 shows three objects from the dataset with the best grasp for each of the objects. The score distribution is shown in Fig. 2, where it is evident that the dataset is biased towards lower scores.

We represent each object as a signed distance field (SDF) rather than using the raw triangle meshes from the provided

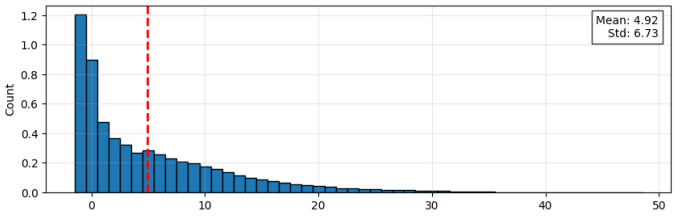


Fig. 2. The score distribution over all samples. The count is in millions.

`.obj` files, as meshes are not directly suitable for neural network processing due to their irregular sampling and varying topology. In contrast, SDFs provide a dense, spatially coherent $48 \times 48 \times 48$ voxel grid, which serves as a uniform and easily digestible input for 3D convolutional architectures. This representation has two main advantages: (1) it avoids the irregular sampling issues inherent to mesh data, and (2) it enables efficient, built-in collision detection during grasp evaluation. To generate the SDFs, we normalize all meshes into a shared coordinate frame by centering them at the origin (recording the translation for later hand-pose correction) and applying a single global scaling factor such that the largest object tightly fits into the cube $[-1, 1]^3$. The normalized meshes are then rasterized into 48^3 SDF grids using `mesh2sdf`. Many meshes in the dataset are not watertight, which can result in non-negative SDF values throughout the grid. Fig. 3 illustrates this effect on two z-slices from the SDF grids from the dataset, where the absence of negative values inside the surface is caused by non-watertight meshes. Section VI discusses the implications in more detail.

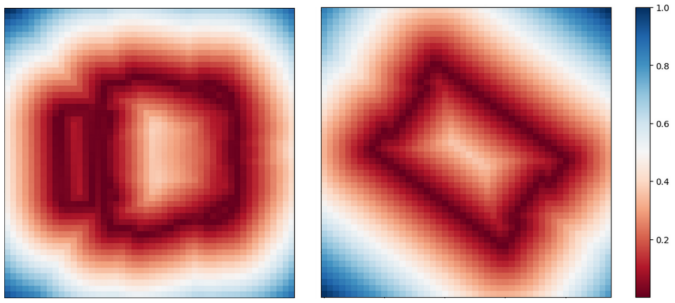


Fig. 3. The z-slices of the SDF voxel grids of two objects. The different SDF values are coloured with warmer colours for smaller SDF values. Note that the smallest SDF values are not negative here.

IV. Method

A. Grasp Quality Network

Given an object represented as an SDF volume $\mathbf{S} \in \mathbb{R}^{48\times48\times48}$ and a grasp configuration $\mathbf{g} \in \mathbb{R}^{19}$, our goal is to learn a function

$$\hat{q} = f_\theta(\mathbf{S}, \mathbf{g}) \tag{1}$$

that predicts the grasp quality score $q \in \mathbb{R}$ for the given pair. The network consists of two main components: (1) an SDF encoder that extracts features from the voxelized object geometry, and (2) a grasp quality prediction head that combines these features with the grasp parameters to output a scalar score. The precise design of these components, including the number of layers, feature dimensions, and activation functions, is the subject of our architecture search described in Section V.

We train the network using mean squared error (MSE) loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\big(f_\theta(\mathbf{S}_i, \mathbf{g}_i) - q_i\big)^2, \tag{2}$$

where N is the number of training samples.

While the ϵ-quality metric can exhibit discontinuities near loss-of-contact events, our objective is not necessarily to minimize absolute error, but to produce a ranking over grasp candidates that is sufficiently consistent to guide grasp optimization (cf. Section V). Our working hypothesis is that by minimizing the MSE loss, the network learns to assign higher scores to higher-quality grasps, thereby inducing the correct ordering in its predictions.

B. Online Optimization Objective

Given a test object SDF $\mathbf{S}$, we seek a grasp $\mathbf{g}$ maximizing a composite objective

$$J(\mathbf{g}\,;\,\mathbf{S}) = f_\theta(\mathbf{S}, \mathbf{g}) \;+\; \lambda \log p_\phi(\mathbf{g}) \;-\; \mu\, \mathcal{L}_{\text{kin}}(\mathbf{g}; \mathbf{S}) \tag{3}$$

where $\log p_\phi(\mathbf{g})$ is a normalizing-flow prior trained on 1M grasp configurations sampled from the dataset and $\mathcal{L}_{\text{kin}}$ a differentiable kinematics loss encouraging fingertip contact and penalizing collisions.

a) Normalizing-flow grasp prior.: To keep optimization within the manifold of plausible grasps, we train a normalizing-flow prior (MAF) p_ϕ, trained on 1M grasps. This prior is *object-agnostic* but captures feasible joint couplings and common poses in 19-D. The term $\lambda \log p_\phi(\mathbf{g})$ is added to discourage implausible grasps.

b) Kinematics loss via SDF.: Let $\mathcal{F}$ be fingertip points and $\mathcal{C}$ collision probe points (hand base, finger roots, and 5 points in each finger), both differentiable functions of $\mathbf{g}$ via forward kinematics. Using trilinear SDF sampling, we define

$$\mathcal{L}_{\text{kin}}(\mathbf{g}; \mathbf{S}) = \underbrace{\sum_{u\in\mathcal{F}} d(\mathbf{S}, u(\mathbf{g}))^2}_{\text{fingertip-to-surface attraction}} + \underbrace{\sum_{v\in\mathcal{C}} \big[\min(0, \mathbf{S}(v(\mathbf{g})))\big]^2}_{\text{collision penalty}}, \tag{4}$$

where $d(\mathbf{S}, x)$ is the signed distance magnitude to the nearest surface (zero-level set). The first term encourages small distances (contact), the second penalizes negative SDF values (interior).[1] We optimize J by gradient ascent on $\mathbf{g}$, holding the SDF encoder fixed per object.

V. Experiments

A. Grasp Quality Predictor

We investigated a range of architectures for predicting grasp quality from an SDF volume $\mathbf{S}$ and grasp parameters $\mathbf{g}$, with the goal of minimizing test-set MSE while maintaining relatively low inference latency.

[1] In practice, some meshes lacked negative interior due to leaks, weakening the collision term; see Sec. VI.

a) Architectures explored.: The baseline model consisted of a 4-layer 3D CNN SDF encoder fused with $\mathbf{g}$ via a multi-layer perceptron (MLP) grasp predictor head. We evaluated several larger-capacity variants with increased convolutional channels and deeper MLPs, as well as smaller-capacity models designed for faster inference. Additional experiments incorporated Spatial Attention, Channel Attention, and hybrid attention mechanisms into the encoder, as well as residual block variants and autoencoder-style encoders trained with reconstruction loss for regularization. While some heavier models marginally improved training loss, the simple 4-layer CNN + MLP design consistently outperformed in validation MSE. Fig. 4 shows the network architecture that resulted in the lowest MSE validation loss.

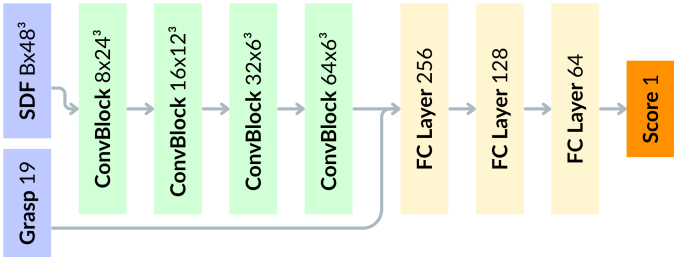


Fig. 4. This is the basic network architecture. It shows the SDF encoder part in green and the grasp quality predictor head in yellow.

b) Training.: We trained all models with the Adam optimizer, using a linearly decreasing learning rate from 1×10^{-3} to 5×10^{-5} over 30 to 100 epochs, batch size 32 to 1024, and early stopping on validation MSE. A batch size of 128 seemed to work best. Training set sizes of 100k, 1M, 5M grasp samples were tested; models trained on larger datasets generally achieved lower validation loss. Batch normalization and more aggressively decreasing LR schedules improved performance consistently, while L^2 weight decay did not yield similar gains.



Fig. 5. The loss curve of the first significant run in red and the last significant run in blue.

c) Results.: Our final model achieved an MSE of **6.08** on the test set (initial baseline: **13.44**). Despite discontinuities in the ϵ-quality metric, this model produced substantially improved grasp ranking quality. Table I summarizes validation loss for representative configurations.

Model Config	**Val Loss**
Scene batches	13.44
Random batches, 1M	11.52
BatchNorm	10.07
Smaller NN, 5M, LR Sched.	8.96
Bigger NN	6.08

TABLE I
MAJOR MODEL CONFIGURATIONS AND THEIR VALIDATION LOSS.

Figure 6 illustrates the predicted grasp quality scores versus the ground-truth ϵ-quality values for all grasps of a single test object. While absolute prediction errors can be non-negligible, the relative ordering of grasps is mostly preserved. This supports our hypothesis that minimizing MSE is sufficient to induce a ranking that guides the subsequent optimization stage effectively. Moreover, the figure highlights that MSE alone may not be the most reliable metric for comparing models, as even a relatively high MSE (14.8 for this object) can still correspond to a well-preserved ordering that is beneficial for optimization. Alternative evaluation metrics such as Spearman's rank correlation or Kendall's tau could better capture this ordering quality; these options are discussed further in Section VI.

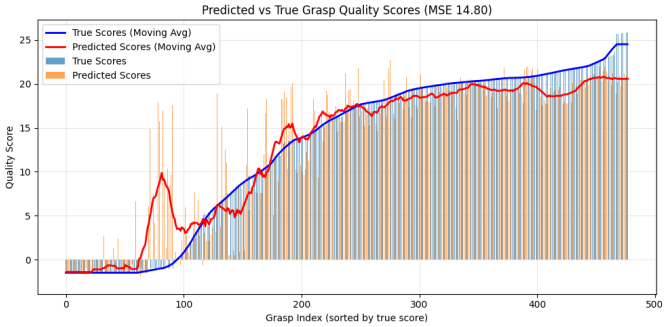


Fig. 6. The ordering of predictions and targets for one object.

B. Online Optimization Results

At test time, we maximize Eq. 3 for a fixed object SDF $\mathbf{S}$ by optimizing the 19-D grasp $\mathbf{g}$.

a) Naïve ascent (quality only).: Ascending only the predicted quality f_θ frequently diverged: parameters grew without bound and contacts became invalid (Fig. 7). Adding ℓ_2 regularization limited parameter magnitudes but still yielded implausible configurations. We attribute this to the optimizer leaving the data manifold, effectively creating adversarial grasps to which f_θ assigns spuriously high scores.

b) Grasp prior.: To keep optimization in-distribution, we trained an object-agnostic Masked Autoregressive Flow prior p_ϕ (5 layers, 64 hidden features) on 1M grasp samples (Adam, learning rate 1×10^{-4}, batch size 128; early stopping at 39/100 epochs). The prior captures joint couplings and typical poses in the 19-D space, and the $\lambda \log p_\phi(\mathbf{g})$ term steers the search toward feasible regions. Incorporating the prior substantially improved convergence and grasp validity (Figs. 8 and 10).

c) Kinematics loss.: Even with the prior, optimized grasps could penetrate the object and lacked reliable fingertip contact. Adding the SDF-based kinematics term $\mathcal{L}_{\text{kin}}$ reduced interpenetration and increased fingertip contact rates; in many runs the kinematics loss approached zero at convergence (Fig. 9).

d) Performance.: For each test object, we pre-encode $\mathbf{S}$ once, initialize $\mathbf{g}$ from prior samples, and launch k parallel Adam runs for 1000 steps (step size 1×10^{-2}) with $(\lambda, \mu) = (0.75, 200)$. On an *M3 MacBook Air* with 24 GB RAM this requires $\sim$3 s per object, and most runs converge within 100–200 steps (< 1 s).

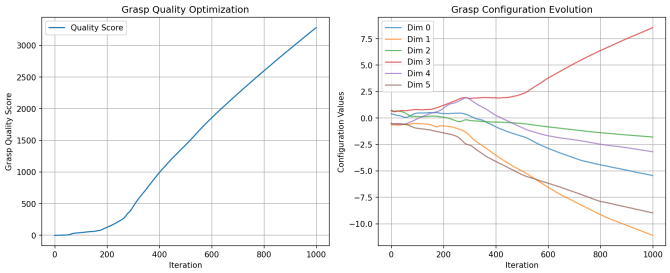


Fig. 7. Maximizing predicted quality alone leads to divergence and invalid contacts.

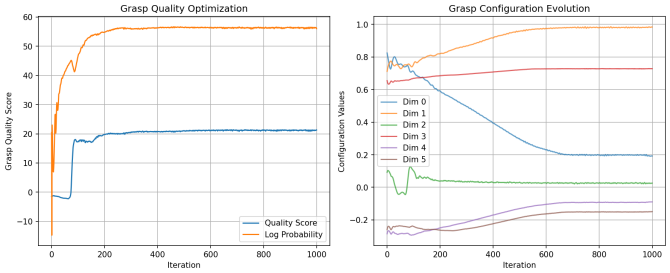


Fig. 8. Adding the normalizing-flow prior yields convergence and generally well-formed grasps.

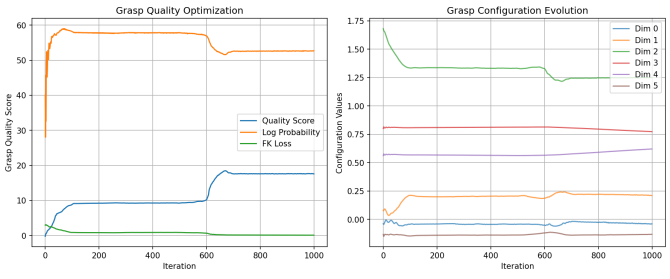


Fig. 9. With both the flow prior and kinematics term, optimization converges to valid grasps and $\mathcal{L}_{\text{kin}}$ nearly vanishes.

e) Qualitative analysis.: With the full objective, solutions typically approach the object with multiple fingertip contacts and minimal interpenetration. Failures occur on thin structures, where sparse collision probes miss intersections, and on leaky meshes whose SDF lacks a negative interior, nullifying the collision penalty. In the latter case, if an initialization places a probe inside the (non-negative) interior, the optimizer can become trapped.

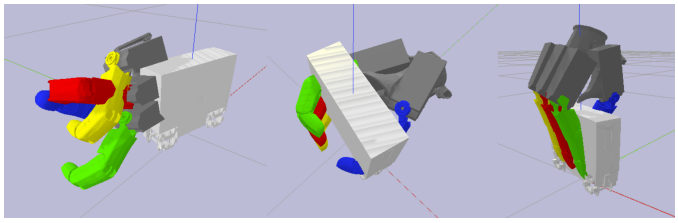


Fig. 10. (1) Quality-only optimization exploits out-of-distribution regions, producing implausible grasps with overestimated scores. (2) The flow prior keeps the search in-distribution, yielding a plausible grasp. (3) The kinematics term further enforces fingertip contact, producing a stable grasp.

VI. Limitations and Future Work

Watertightness. Mesh leaks often eliminated negative SDF interiors, weakening collision penalties and, in some cases, even preventing convergence. Pre-filling, watertight remeshing, or robust inside-outside classifiers could restore this signal.

Sparse collision probes. With only 21 points (bases, roots, a few joints), thin structures can be missed. Increasing probe density and adding palm/phalange surfaces would help.

Decoupling hand-pose and grasp. In some cases, we observed suboptimal hand pose updates during online adaptation, in particular concerning the hand rotation. We attribute this to rotational bias in the flow term. Decoupling the hand-pose from the prior and using a separate objective term, e.g. ℓ^2-regularization, may further improve results.

Object-agnostic prior. The flow prior ignores object context; conditioning $p_\phi(\mathbf{g} \mid \mathbf{S})$ could further bias the search toward object-compatible grasps. Simultaneously, this may eliminate the aforementioned rotational bias.

Loss for ordering. Incorporating a ranking loss or a margin around the contact manifold may improve optimization behavior without overfitting score magnitudes.

Dynamics. Our evaluation is geometric and quasi-static; real execution will have incomplete information and often dynamic geometry.

VII. Conclusion

We presented a simple and effective approach to online grasp adaptation. A compact SDF-based predictor supplies a smooth surrogate for ϵ-quality, while a flow-based grasp prior and an SDF-driven kinematics loss regularize the optimizer toward valid, contact-rich grasps. Despite discontinuous ground-truth scores, this combination reliably improves both validity and quality of optimized grasps. Addressing mesh watertightness and enriching collision/contact modeling are promising next steps.

a) Source Code: https://github.com/hannesill/grasp-quality

REFERENCES

[1] J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, X. Liu, J. A. Ojea, and K. Goldberg, "Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics," in *Robotics: Science and Systems (RSS)*, 2017. [Online]. Available: https://arxiv.org/abs/1703.09312

[2] M. V. der Merwe, Q. Lu, B. Sundaralingam, M. Matak, and T. Hermans, "Learning continuous 3d reconstructions for geometrically aware grasping," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 11 516–11 522. [Online]. Available: https://arxiv.org/abs/1910.00983

[3] M. Liu, Z. Pan, K. Xu, K. Ganguly, and D. Manocha, "Deep differentiable grasp planner for high-dof grippers," in *Proceedings of Robotics: Science and Systems (RSS)*, 2020. [Online]. Available: https://roboticsproceedings.org/rss16/p066.pdf

[4] D. Winkelbauer, B. Bäuml, M. Humt, N. Thuerey, and R. Triebel, "A two-stage learning architecture that generates high-quality grasps for a multi-fingered hand," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Kyoto, Japan, 2022. [Online]. Available: https://elib.dlr.de/191780/1/winkelbauer2022.pdf

[5] J. J. Park, P. Florence, J. Straub, R. Newcombe, and S. Lovegrove, "Deepsdf: Learning continuous signed distance functions for shape representation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/papers/Park_DeepSDF_Learning_Continuous_Signed_Distance_Functions_for_Shape_Representation__2019_paper.pdf